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MCQs for CBSE Class 12 Biology Chapter 1: Sexual Reproduction in Flowering Plants

1. What is the male reproductive organ of a flower?
a) Stigma
b) Anther
c) Ovary
d) Style
Answer: b) Anther

2. The process of transfer of pollen grains from the anther to the stigma of a flower is called:
a) Fertilisation
b) Pollination
c) Germination
d) Ovulation
Answer: b) Pollination

3. Which of the following is not a method of self-pollination in flowers?
a) Cleistogamy
b) Wind pollination
c) Geitonogamy
d) Autogamy
Answer: b) Wind pollination

4. In which part of the flower's pistil does fertilisation occur?
a) Ovary
b) Style
c) Stigma
d) Ovule
Answer: a) Ovary

5. The fusion of male and female gametes during sexual reproduction in flowering plants leads to the formation of:
a) Embryo
b) Zygote
c) Endosperm
d) Seed
Answer: b) Zygote

6. In angiosperms, meiosis occurs during the formation of:
a) Pollen grains

b) Ovules
c) Endosperm
d) Embryo sac
Answer: d) Embryo sac

7. What is the purpose of the pollen tube in plants?
a) To provide support to the pollen grain
b) To protect the pollen grain during transportation
c) To help in fertilisation by carrying male gametes to the ovary
d) To store food for the developing embryo
Answer: c) To help in fertilisation by carrying male gametes to the ovary

8. The phenomenon where a single pollen grain germinates and produces two male gametes is called:
a) Binary fission
b) Double fertilisation
c) Bipolar division
d) Triple fusion
Answer: b) Double fertilisation

9. Which part of the embryo sac nourishes the developing embryo?
a) Synergids
b) Antipodal cells
c) Egg cell
d) Central cell
Answer: d) Central cell

10. Which hormone is responsible for the development of fruits from the ovary after fertilisation?
a) Auxin
b) Cytokinin
c) Gibberellin
d) Ethylene
Answer: d) Ethylene

11. The transfer of pollen grains from the anther to the stigma of the same flower is known as:
a) Geitonogamy
b) Cross-pollination
c) Cleistogamy
d) Autogamy
Answer: d) Autogamy

12. Which of the following is not a reproductive part of a flower?
a) Sepal
b) Petal

c) Stamen
d) Style
Answer: a) Sepal

13. In heterogamous flowers, the male and female reproductive organs mature:
a) At the same time
b) On different flowers of the same plant
c) On different plants of the same species
d) In different seasons
Answer: c) On different plants of the same species

14. The edible part of the mango fruit is the:
a) Mesocarp
b) Epicarp
c) Endocarp
d) Seed
Answer: a) Mesocarp

15 Which of the following is a non-endospermic seed?

A) Corn
B) Pea
C) Coconut
D) Sunflower
Answer: B) Pea

MCQs for CBSE Class 12 Biology Chapter 2: Human Reproduction

1. Which of these is not a part of the female reproductive system?

a) Ovary

b) Fallopian tube

c) Uterus

d) Prostate gland

Answer: d) Prostate gland

2. The primary sex organ(s) in males is/are:

a) Testes

b) Epididymis

c) Vas deferens

d) Seminal vesicles

Answer: a) Testes

3. The process of formation of mature sperm cells is called:

a) Oogenesis

b) Ovulation

c) Spermatogenesis

d) Menstruation

Answer: c) Spermatogenesis

4. Fertilisation in humans occurs in the:

a) Uterus

b) Vagina

c) Ovary

d) Fallopian tube

Answer: d) Fallopian tube

5. The hormone responsible for the development of secondary sexual characteristics in males is:

a) Estrogen

b) Progesterone

c) Testosterone

d) Follicle-stimulating hormone (FSH)
Answer: c) Testosterone

6. The menstrual cycle in females is regulated by the hormones:
a) Estrogen and progesterone
b) Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)
c) Luteinizing hormone (LH) and testosterone
d) Prolactin and oxytocin
Answer: b) Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)

7. The release of a mature egg from the ovary is known as:
a) Menstruation
b) Fertilisation
c) Ovulation
d) Implantation
Answer: c) Ovulation

8. The inner lining of the uterus, where the fertilised egg implants and develops, is called the:
a) Endometrium
b) Myometrium
c) Perimetrium
d) Cervix
Answer: a) Endometrium

9. The hormone responsible for milk production in mammary glands is:
a) Estrogen
b) Progesterone
c) Prolactin
d) Oxytocin
Answer: c) Prolactin

10. The process of shedding of the uterine lining along with blood is known as:
a) Ovulation
b) Fertilisation
c) Menstruation
d) Implantation
Answer: c) Menstruation

11. The maximum number of spermatozoa are stored in the:

a) Epididymis

b) Seminal vesicles

c) Vas deferens

d) Prostate gland

Answer: a) Epididymis

12. In humans, fertilisation typically occurs in the:

a) Uterus

b) Cervix

c) Vagina

d) Fallopian tube

Answer: d) Fallopian tube

13. The hormone responsible for the contraction of uterine muscles during childbirth is:

a) Estrogen

b) Progesterone

c) Prolactin

d) Oxytocin

Answer: d) Oxytocin

14. Which of the following is a STI (sexually transmitted infection)?

a) Hepatitis B

b) Influenza

c) Tuberculosis

d) Malaria

Answer: a) Hepatitis B

15. The surgical method of permanent contraception in males is called:

a) Vasectomy

b) Tubal ligation

c) Hysterectomy

d) Oophorectomy

Answer: a) Vasectomy

MCQs for CBSE Class 12 Biology Chapter 3: Reproductive Health

1. Which of the following is a barrier method of contraception?

a) Tubectomy

b) Oral contraceptives

c) Condom

d) Copper-T

Answer: c) Condom

2. Emergency contraception is effective if used within:

a) 12 hours

b) 24 hours

c) 48 hours

d) 72 hours

Answer: d) 72 hours

3. Which STI is caused by the bacterium *Neisseria gonorrhoeae*?

a) Syphilis

b) Genital herpes

c) Gonorrhea

d) HIV/AIDS

Answer: c) Gonorrhea

4. In which part of the female reproductive system does fertilisation occur?

a) Uterus

b) Vagina

c) Fallopian tubes

d) Cervix

Answer: c) Fallopian tubes

5. The surgical method of permanent contraception in females is called:

a) Tubectomy

b) Vasectomy

c) IUD insertion

d) Oral contraception

Answer: a) Tubectomy

6. Which contraceptive method works by inhibiting ovulation?

a) Condom

b) Oral contraceptive pill

c) Copper-T

d) Diaphragm

Answer: b) Oral contraceptive pill

7. Which of these is not a type of assisted reproductive technology (ART)?

a) In vitro fertilisation (IVF)

b) Intrauterine insemination (IUI)

c) Surrogacy

d) Hysterectomy

Answer: d) Hysterectomy

8. Which of these is a non-hormonal contraceptive method?

a) Oral contraceptive pill

b) Copper-T

c) Contraceptive patch

d) Emergency contraception

Answer: b) Copper-T

9. Which STI is caused by the human immunodeficiency virus (HIV)?

a) Chlamydia

b) Gonorrhea

c) Syphilis

d) HIV/AIDS

Answer: d) HIV/AIDS

10. Which of the following is a permanent method of contraception for males?

a) Tubectomy

b) Vasectomy

c) Intrauterine device (IUD)

d) Contraceptive implant

Answer: b) Vasectomy

11. The method of contraception that is reversible and involves blocking or sealing the fallopian tubes is called:

a) Condom

b) IUD insertion

c) Tubectomy

d) Laparoscopy

Answer: c) Tubectomy

12. Which STI is caused by the bacterium Treponema pallidum?

a) Genital herpes

b) Gonorrhea

c) Syphilis

d) Chlamydia

Answer: c) Syphilis

13. Which of the following is not a barrier method of contraception?

a) Diaphragm

b) Condom

c) Copper-T

d) Cervical cap

Answer: c) Copper-T

14. Which method of contraception is known as the “morning-after pill”?

a) Oral contraceptive pill

b) Copper-T

c) Emergency contraception

d) Sterilization

Answer: c) Emergency contraception

15. Which contraceptive method provides protection against both unwanted pregnancies and sexually transmitted infections?

a) Oral contraceptive pill

b) Barrier methods (e.g., condom)

c) Intrauterine device (IUD)

d) Sterilization

Answer: b) Barrier methods (e.g., condom)

MCQs for CBSE Class 12 Biology Chapter 4: Principles of Inheritance and Variations

1. In Mendel's experiments on pea plants, the trait that disappeared in the F1 generation but reappeared in the F2 generation is called:

a) Dominant trait

b) Recessive trait

c) Co-dominant trait

d) Linked trait

Answer: b) Recessive trait

2. The term "homozygous" refers to an individual that has:

a) Two different alleles for a particular gene

b) Two identical alleles for a particular gene

c) Multiple alleles for a particular gene

d) No alleles for a particular gene

Answer: b) Two identical alleles for a particular gene

3. The genetic makeup of an individual is referred to as its:

a) Phenotype

b) Genotype

c) Allele

d) Homozygosity

Answer: b) Genotype

4. The observed physical characteristics of an individual are referred to as its:

a) Genotype

b) Phenotype

c) Allele

d) Dominance

Answer: b) Phenotype

5. Which of the following is an example of a sex-linked inheritance?

a) Blood type

b) Eye color

c) Height

d) Color blindness

Answer: d) Color blindness

6. The inheritance of traits controlled by a single gene with two or more alleles is called:

a) Incomplete dominance

b) Co-dominance

c) Multiple alleles

d) Polygenic inheritance

Answer: c) Multiple alleles

7. A gene located on the X-chromosome is called a/an:

a) Autosomal gene

b) Recessive gene

c) Dominant gene

d) Sex-linked gene

Answer: d) Sex-linked gene

8. In a dihybrid cross, if the two genes assort independently, the phenotypic ratio in the F2 generation will be:

a) 3:1

b) 1:2:1

c) 9:3:3:1

d) 1:1

Answer: c) 9:3:3:1

9. The phenomenon where a single gene influences multiple traits is called:

a) Incomplete dominance

b) Co-dominance

c) Pleiotropy

d) Polygenic inheritance

Answer: c) Pleiotropy

10. In humans, the inheritance of ABO blood groups follows the pattern of:

a) Complete dominance

b) Incomplete dominance

c) Co-dominance

d) Multiple alleles

Answer: c) Co-dominance

11. The inheritance of skin color in humans is an example of:

a) Complete dominance

b) Incomplete dominance

c) Co-dominance

d) Polygenic inheritance

Answer: d) Polygenic inheritance

12. The presence of an additional copy of a chromosome in a cell is known as:

a) Polyploidy

b) Aneuploidy

c) Deletion

d) Duplication

Answer: b) Aneuploidy

13. Down syndrome, caused by an extra copy of chromosome 21, is an example of:

a) Polyploidy

b) Aneuploidy

c) Deletion

d) Duplication

Answer: b) Aneuploidy

14. The process of exchange of genetic material between homologous chromosomes during meiosis is called:

a) Crossing over

b) Inversion

c) Translocation

d) Nondisjunction

Answer: a) Crossing over

MCQs for CBSE Class 12 Biology Chapter 5: Molecular Basis of Inheritance

1. A short piece of DNA, having 20 base pairs, was analyzed to find the number of nucleotide bases in each of the polynucleotide strands. Some of the results are shown in the table.

	Number of nucleotide bases			
	Adenine	**Cytosine**	**Guanine**	**Thymine**
Strand 1	**4**	**4**		
Strand 2		**5**		

How many nucleotides containing Adenine were present in strand 2?
a) 2
b) 4
c) 5
d) 7
Answer: d)7

2. What is the smallest part of a DNA molecule that can be changed by a point mutation?
a) Oligonucleotide
b) Codon
c) Gene
d) Nucleotide
Answer: d) Nucleotide

3. In bacteria, the transfer of genetic material from one bacterium to another through direct cell-to-cell contact is called:
a) Transformation
b) Transduction
c) Conjugation
d) Replication
Answer: c) Conjugation

4. The genetic code is said to be degenerate because:
a) Each amino acid has multiple codons
b) Each codon codes for multiple amino acids
c) The start codon codes for multiple amino acids
d) The stop codon codes for multiple amino acids
Answer: a) Each amino acid has multiple codons

5. Which of the following is an example of a silent mutation?
a) Missense mutation
b) Nonsense mutation
c) Frame-shift mutation
d) None of the above

Answer: d) None of the above

6. The enzyme responsible for adding telomeres at the ends of eukaryotic chromosomes is:
a) DNA polymerase
b) Primase
c) Telomerase
d) Ligase
Answer: c) Telomerase

7. The Central Dogma of molecular biology states that:
a) DNA is transcribed into RNA, which is then translated into proteins.
b) RNA is transcribed into DNA, which is then translated into proteins.
c) Proteins are transcribed into DNA, which is then translated into RNA.
d) Proteins are transcribed into RNA, which is then translated into DNA.
Answer: a) DNA is transcribed into RNA, which is then translated into proteins.

8. Which of the following is not a type of RNA involved in protein synthesis?
a) Messenger RNA (mRNA)
b) Transfer RNA (tRNA)
c) Ribosomal RNA (rRNA)
d) Regulatory RNA (regRNA)
Answer: d) Regulatory RNA (regRNA)

9. The genetic material in viruses can be:
a) DNA only
b) RNA only
c) Either DNA or RNA
d) Proteins
Answer: c) Either DNA or RNA

10. DNA replication is said to be semiconservative because:
a) One strand of DNA is newly synthesised, while the other strand is conserved.
b) Both strands of DNA are newly synthesised.
c) Both strands of DNA are conserved.
d) One strand of DNA is degraded, while the other strand is newly synthesised.
Answer: a) One strand of DNA is newly synthesised, while the other strand is conserved.

CBSE Class 12 Biology
Chapter 6 Evolution MCQs

1. Which scientist proposed the concept of the inheritance of acquired traits?
a) Charles Darwin
b) Jean-Baptiste Lamarck
c) Gregor Mendel
d) Alfred Russel Wallace
Answer: b) Jean-Baptiste Lamarck

2. The term "homologous structures" refers to structures that:
a) Have similar functions but different evolutionary origins.
b) Have similar structures but different functions.
c) Have both similar structures and functions.
d) Have neither similar structures nor functions.
Answer: c) Have both similar structures and functions.

3. According to the Hardy-Weinberg principle, the frequency of alleles in a population will remain constant if:
a) Mutation occurs at a high rate.
b) Genetic drift is absent.
c) Natural selection is intense.
d) Gene flow is restricted.
Answer: b) Genetic drift is absent.

4. The extinction of dinosaurs is associated with:
a) Volcanic eruptions
b) Asteroid impact
c) Climate change
d) Plate tectonics
Answer: b) Asteroid impact

5. The phenomenon where unrelated species evolve similar traits due to similar environmental pressures is called:
a) Convergent evolution
b) Divergent evolution
c) Coevolution
d) Adaptive radiation
Answer: a) Convergent evolution

6. The idea that evolution occurs in small, incremental steps over long periods of time is known as:
a) Gradualism
b) Punctuated equilibrium
c) Speciation
d) Adaptive radiation
Answer: a) Gradualism

7. The evolution of wings in birds and bats is an example of:
a) Convergent evolution
b) Divergent evolution
c) Coevolution
d) Parallel evolution
Answer: a) Convergent evolution

8. The evolutionary theory proposed by Charles Darwin is based on the concept of:
a) Inheritance of acquired traits
b) Natural selection
c) Genetic drift
d) Mutation
Answer: b) Natural selection

9. The process by which new species evolve from a common ancestor is called:
a) Speciation
b) Hybridization
c) Genetic drift
d) Mutation
Answer: a) Speciation

10. The concept of "survival of the fittest" is associated with:
a) Jean-Baptiste Lamarck
b) Charles Darwin
c) Alfred Russel Wallace
d) Gregor Mendel
Answer: b) Charles Darwin

CBSE Class 12 Biology Chapter 7 Health and Disease MCQs

1. **Which of the following is not a viral disease?**
a) Tuberculosis
b) Influenza
c) Measles
d) Hepatitis B
Answer: a) Tuberculosis

2. **Antibiotics are effective against:**
a) Viruses
b) Bacteria
c) Fungi
d) Protozoans
Answer: b) Bacteria

3. **The causative agent of malaria is a:**
a) Virus
b) Bacterium
c) Fungus
d) Protozoan
Answer: d) Protozoan

4. **The immune response that occurs immediately upon exposure to an antigen is called:**
a) Cell-mediated immunity
b) Active immunity
c) Passive immunity
d) Innate immunity
Answer: d) Innate immunity

5. **The human immunodeficiency virus (HIV) primarily attacks and destroys:**
a) Red blood cells
b) White blood cells
c) Platelets
d) Neurons
Answer: b) White blood cells

6. **Which of the following is an autoimmune disease?**
a) Diabetes
b) Tuberculosis
c) Malaria
d) Influenza
Answer: a) Diabetes

7. **Vaccines work by:**
a) Stimulating the production of antibodies
b) Killing the pathogen directly
c) Reducing inflammation
d) Blocking the entry of pathogens into the body
Answer: a) Stimulating the production of antibodies

8. **The transmission of a disease from an infected mother to her unborn child is called:**
a) Vertical transmission
b) Horizontal transmission
c) Zoonotic transmission
d) Mechanical transmission
Answer: a) Vertical transmission

9. **The causative agent of dengue fever is transmitted by:**
a) Mosquitoes
b) Ticks
c) Fleas
d) Lice
Answer: a) Mosquitoes

10. **The Widal test is used for the diagnosis of which of the following diseases?**
a) Malaria
b) Tuberculosis
c) Dengue fever
d) Typhoid fever
Answer: d) Typhoid fever

CBSE Class 12 Biology Chapter 8 Microbes in Human Welfare MCQs

1. Microbes used in the production of antibiotics belong to which group?
a) Fungi
b) Bacteria
c) Viruses
d) Protozoans
Answer: b) Bacteria

2. Which microorganism is commonly used in the production of alcohol through fermentation?
a) *Saccharomyces cerevisiae*
b) *Aspergillus niger*
c) *Penicillium notatum*
d) *Escherichia coli*
Answer: a) *Saccharomyces cerevisiae*

3. Which enzyme is produced by microorganisms to convert lactose into lactic acid during the production of curd?
a) Protease
b) Amylase
c) Lactase
d) Lactic acidase
Answer: c) Lactase

4. Microbes are used in the production of biogas through the process of:
a) Fermentation
b) Photosynthesis
c) Respiration
d) Transpiration
Answer: a) Fermentation

5. The process of using microorganisms to detoxify or remove pollutants from the environment is called:
a) Bioremediation
b) Bioaccumulation
c) Biomagnification
d) Bioprocessing
Answer: a) Bioremediation

6. Which microorganism is used in the production of single-cell protein (SCP)?
a) Escherichia coli
b) Aspergillus niger
c) Rhizobium
d) Spirulina
Answer: d) Spirulina

7. Which microorganism is responsible for the production of citric acid in the industry?
a) *Aspergillus niger*
b) *Saccharomyces cerevisiae*
c) *Rhizobium*

d) *Lactobacillus*
Answer: a) *Aspergillus niger*

8. Microbes used in the process of retting of fibres are:
a) Bacteria
b) Fungi
c) Protozoans
d) Viruses
Answer: b) Fungi

9. The microorganism used in the production of bread and beer is:
a) *Saccharomyces cerevisiae*
b) Escherichia coli
c) Aspergillus niger
d) Lactobacillus
Answer: a) *Saccharomyces cerevisiae*

10. Which microorganism is used in the biological control of insect pests?
a) *Bacillus thuringiensis*
b) Rhizobium
c) Azotobacter
d) Pseudomonas
Answer: a) *Bacillus thuringiensis*

CBSE Class 12 Biology Chapter 9 Biotechnology: Principles and Processes MCQs

1. **Which of the following is a process used in biotechnology to transfer genes into a host organism?**
a) PCR (Polymerase Chain Reaction)
b) Gel Electrophoresis
c) Cloning
d) Recombinant DNA technology
Answer: d) Recombinant DNA technology

2. **The enzyme responsible for cutting DNA at specific sequences in recombinant DNA technology is called:**
a) DNA ligase
b) DNA polymerase
c) Restriction enzyme
d) RNA polymerase
Answer: c) Restriction enzyme

3. **What is the function of a selectable marker in recombinant DNA technology?**
a) To differentiate between transformed and non-transformed cells
b) To help in DNA replication
c) To amplify the DNA
d) To join DNA fragments together
Answer: a) To differentiate between transformed and non-transformed cells

4. **The process of separating DNA fragments based on their size and charge using an electric field is called:**
a) PCR (Polymerase Chain Reaction)
b) DNA sequencing
c) Gel electrophoresis
d) DNA replication
Answer: c) Gel electrophoresis

5. **The enzyme used to produce complementary DNA (cDNA) from an mRNA template is:**
a) DNA polymerase
b) Reverse transcriptase
c) DNA ligase
d) Restriction enzyme
Answer: b) Reverse transcriptase

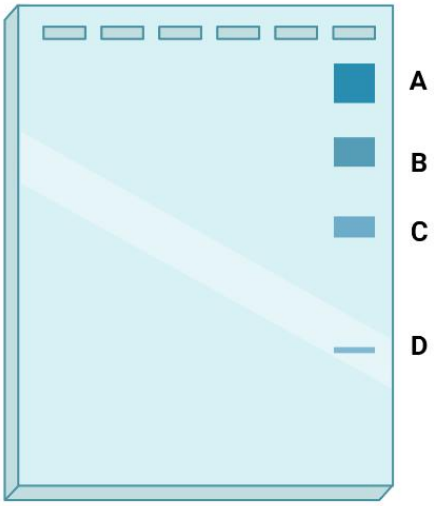
6. **The technique used to amplify a specific region of DNA to produce multiple copies is called:**
a) Recombinant DNA technology
b) DNA sequencing
c) PCR (Polymerase Chain Reaction)
d) Gel electrophoresis
Answer: c) PCR (Polymerase Chain Reaction)

7. **Which restriction site is not present in pBR322 plasmid?**
a) EcoRI
b) Hind III

c) Hind II
d) Sal I
Answer: c) Hind II

8. **pBR322 plasmid is resistant to which of the following antibiotics?**
a) Ampicillin
b) tetracycline
c) Penicillin
d) both a and b
Answer: d) both a and b

9. **Which DNA fragment is the smallest?**



a) A
b) B
c) C
d) D
Answer: d) D

10. **Which of these is the feature of a cloning vector?**
a) Ori site
b) Selectable marker
c) Cloning site
d) All of the above
Answer: d) All of the above

CBSE Class 12 Biology Chapter 10 Biotechnology and its Applications MCQs

1. **Which organisation make decisions regarding the validity of GM research and the safety of introducing GM-organisms**
a) WHO
b) GEAC
c) Government of India
d) Non of the above
Answer: b) GEAC (Genetic Engineering Approval Committee)

2. **Which biotechnological application is used to produce large quantities of insulin for medical use?**
a) DNA fingerprinting
b) Gene therapy
c) PCR (Polymerase Chain Reaction)
d) Recombinant DNA technology
Answer: d) Recombinant DNA technology

3. **Which of the following is an example of a genetically** modified (GM) crop?
a) Rice
b) Wheat
c) Cotton
d) Mustard
Answer: c) Cotton

4. **The method of producing thousands of plants through tissue culture is called**
a) Micro-propagation
b) Vegetative propagation
c) Tissue culturing
d) Non of the above
Answer: a) Micro-propagation

The Full form of GMOs
a) Government Mentioned Organisms
b) Genetically Mature Organisms

c) Genetically Modified Objects
d) Genetically Modified Organisms
Answer: d) Genetically Modified Organisms

Full form of dsRNA
a) Double-stranded ribonucleic acid
b) Double-stranded deoxyribonucleic acid
c) Double-segmented ribonucleic acid
d) None of the above
Answer: a) Double-stranded ribonucleic acid

Which peptide is not present in mature insulin?
a) A peptide
b) B peptide
c) C peptide
d) A and B peptides
Answer: c) C peptide

First gene therapy was given in?
a) 1990
b) 1909
c) 1991
d) 1900
Answer: a) 1990

The correct full form of BT in Bt Cotton is;
a) *Bacillus thuringensis*
b) *Bacillus thurinogiensis*
c) *Bacillus thuringiensis*
d) *Bacillus thoringiensis*
Answer: c) *Bacillus thuringiensis*

This capacity to generate a whole plant from any cell/explant is called
a) Pluripotency
b) Totipotency
d) Both a and b
c) None of these
Answer: b) Totipotency

CBSE Class 12 Biology Chapter 11 Organisms and Populations MCQs

1. **Which of the following is an example of a density-dependent factor that regulates population growth?**
a) Temperature
b) Precipitation
c) Predation
d) Earthquake
Answer: c) Predation

2. **The maximum number of individuals of a species that an ecosystem can support without being degraded is known as:**
a) Carrying capacity
b) Population density
c) Biotic potential
d) Logistic growth
Answer: a) Carrying capacity

3. **The pattern of growth where the population increases rapidly and then stabilises near the carrying capacity is known as:**
a) Exponential growth
b) Logistic growth
c) Geometric growth
d) Boom-and-bust cycle
Answer: b) Logistic growth

4. **Which of the following factors is an abiotic component of an ecosystem?**
a) Predators
b) Parasites
c) Soil
d) Herbivores
Answer: c) Soil

5. **A group of individuals of the same species living in the same area and capable of interbreeding is called:**
a) Community

b) Ecosystem
c) Population
d) Habitat
Answer: c) Population

6. **Which one is commensalism?**

	Species A	Species B
i	+	0
ii	+	+
iii	0	+
iv	-	+

a) ii and iv
b) only i
c) i and iii
d) only iii
Answer: c) i and iii

7. **Maximum growth rate occurs in**
a) Senescent phase
b) Lag phase
c) Exponential phase
d) Stationary phase
Answer: c) Exponential phase

8. **Plant species having a wide range of genetical distribution evolve into a local population known as**
a) Ecotype
b) Biome
c) Ecosystem
d) Population
Answer: a) Ecotype

9. **Geometric representation of age structure is a characteristics of**
a) Population
b) Land scape

c) Ecosystem
d) Biotic community
Answer: a) Population

10. **The tendency of population to remain in genetic equlibrium may be disturbed by**
a) Lack of migration
b) Lack of mutations
c) Lack of random mating
d) Randam mating
Answer: c) Lack of random mating

CBSE Class 12 Biology Chapter 12 Ecosystem MCQs

1. An ecosystem consists of
a) Only living organisms
b) Only non-living components
c) Only plants and animals
d) Both living organisms and their physical environment
Answer: d) Both living organisms and their physical environment

2. The primary source of energy in most ecosystems is
a) Wind energy
b) Solar energy
c) Geothermal energy
d) Biomass energy
Answer: b) Solar energy

3. Decomposers in an ecosystem are responsible for:
a) Capturing solar energy and converting it into food
b) Breaking down dead organic matter into simple compounds
c) Producing oxygen through photosynthesis
d) Absorbing nutrients from the soil for plant growth
Answer: b) Breaking down dead organic matter into simple compounds

4. Which trophic level in an ecosystem represents the primary producers?
a) Herbivores
b) Carnivores
c) Producers
d) Decomposers
Answer: c) Producers

5. The biotic component of an ecosystem includes:
a) Air, water, and soil
b) Producers, consumers, and decomposers
c) Sunlight and minerals
d) All living organisms
Answer: b) Producers, consumers, and decomposers

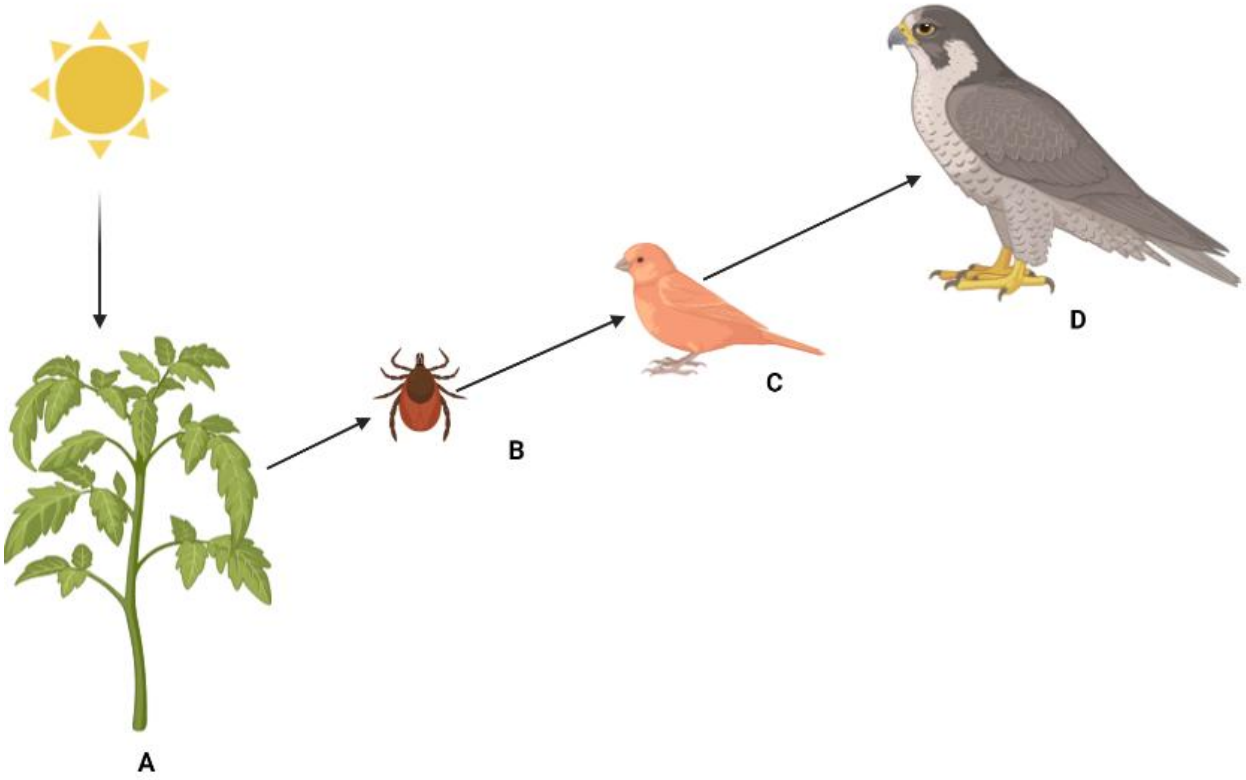
6. The transfer of energy and nutrients between trophic levels in an ecosystem is known as
a) Decomposition
b) Food chain
c) Biomagnification
d) Photosynthesis
Answer: b) Food chain

7. Which of the following statements is true about a food web in an ecosystem?
a) It represents the flow of energy in a single direction from producers to decomposers.
b) It consists of only a single chain of feeding relationships.
c) It shows the interconnectedness of multiple food chains.
d) It includes only primary consumers and decomposers.
Answer: c) It shows the interconnectedness of multiple food chains.

8. Which of the following is an abiotic factor in an ecosystem?
a) Producers
b) Consumers
c) Temperature
d) Decomposers
Answer: c) Temperature

9. The ultimate source of energy in an ecosystem is
a) Producers
b) Consumers
c) Sunlight
d) Decomposers
Answer: c) Sunlight

10. Complete the food chain with the correct terms:



	A	**B**	**C**	**D**
i	Primary Consumer	Producer	Secondary Consumer	Tertiary Consumer
ii	Producer	Secondary Consumer	Primary Consumer	Tertiary Consumer
iii	Producer	Primary Consumer	Secondary Consumer	Tertiary Consumer
iv	Tertiary Consumer	Secondary Consumer	Primary Consumer	Producer

options;

a) i

b) ii

c) iii

d) iv

Answer: c) iii

CBSE Class 12 Biology Chapter 13 Biodiversity and Conservation MCQs

1. Biodiversity refers to:
a) The variety of genetic traits in a population
b) The variety of species and ecosystems in a region
c) The total number of individuals in an ecosystem
d) The variety of physical environments in an area
Answer: b) The variety of species and ecosystems in a region

2. The loss of biodiversity due to human activities is called
a) Evolution
b) Conservation
c) Extinction
d) Biodiversity
Answer: c) Extinction

3. Which of the following is a direct threat to biodiversity?
a) Sustainable development
b) Habitat preservation
c) Pollution
d) Ecotourism
Answer: c) Pollution

4. Which of the following is an in-situ conservation method?
a) Seed banks
b) Botanical gardens
c) National parks
d) Wildlife sanctuaries
Answer: c) National parks

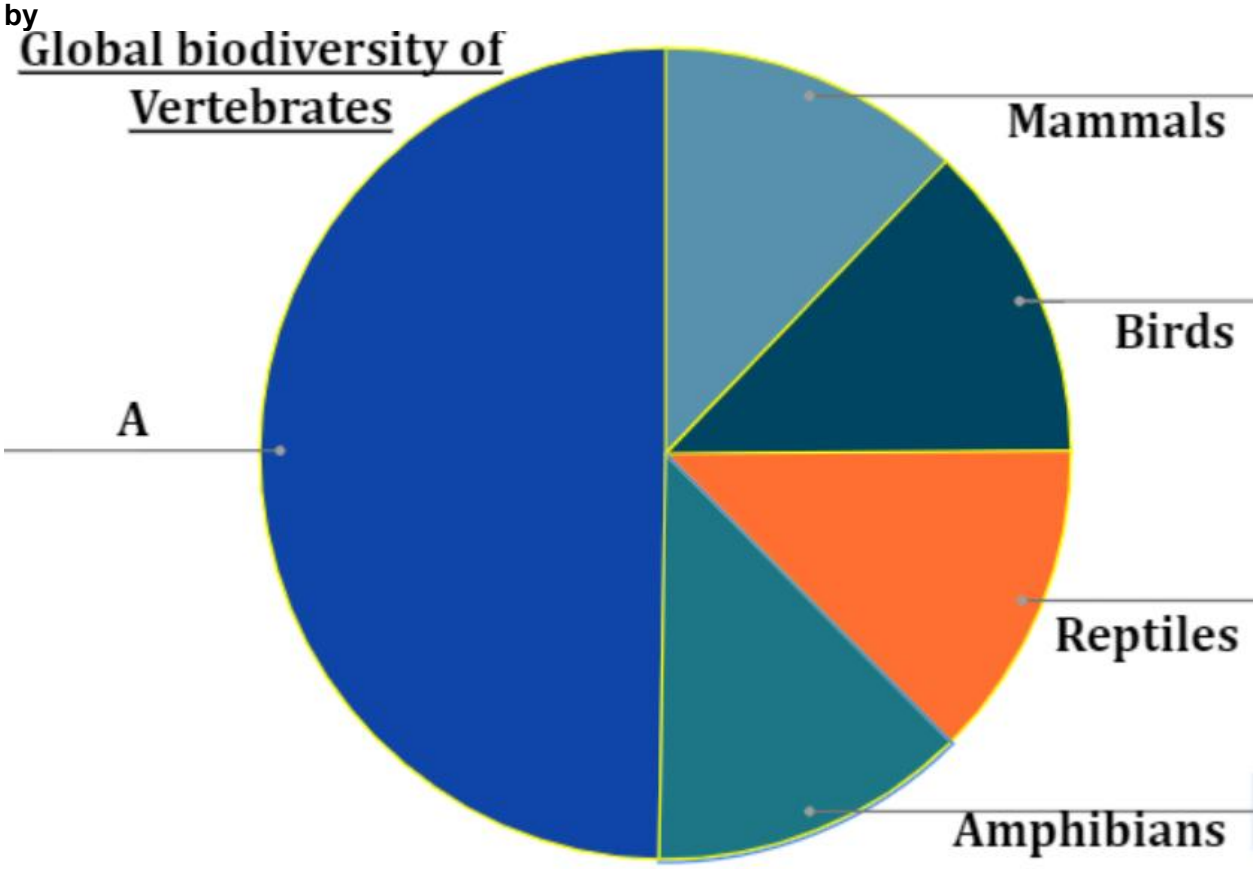
5. Which of the following is an example of an endemic species?
a) Tiger (*Panthera tigris*)
b) Polar bear (*Ursus maritimus*)
c) Komodo dragon (*Varanus komodoensis*)
d) Giant panda (*Ailuropoda melanoleuca*)
Answer: c) Komodo dragon (*Varanus komodoensis*)

6. Which of the following is an example of an ex-situ conservation method?
a) Wildlife sanctuary
b) National Park
c) Zoo
d) Biosphere reserve
Answer: (c) Zoo

7. The loss of a species from a particular area but not from the entire planet is known as
a) Extinction
b) Endemism
c) Biodiversity hotspot
d) Extirpation
Answer: d) Extirpation

8. Biodiversity is the term popularised by the sociobiologist
a) Karl Marx
b) Edward Wilson
c) Herbert Spencer
d) Robert E. Park
Answer: b) Edward Wilson

9. In the global biodiversity pie chart of Vertebrates given below, 'A' is covered by



a) Insects
b) Fishes
c) Angiosperms
d) None of the above
Answer: b) Fishes

10. On a logarithmic scale, the relationship is a straight line described by the equation
a) log S = log C + Z log A
b) log S = log A + Z log C
c) log C = log S + Z log A
d) log Z = log S + C log A
Answer: a) log S = log C + Z log A